

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** Database Management Systems  
**COURSE CODE:** CSA05

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### COURSE OUTCOME COVERED

**CO1:** *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)  
Assessment Tool Weightage: 40%

| QUESTIONS |                                                                                                                  |                  |
|-----------|------------------------------------------------------------------------------------------------------------------|------------------|
| Q. No     | Question                                                                                                         | Bloom's Level    |
| 1         | Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema. | BL1 – Remember   |
| 2         | Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.  | BL2 – Understand |
| 3         | Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.   | BL2 – Understand |
| 4         | Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.              | BL2 – Understand |
| 5         | Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.        | BL1 – Remember   |

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**Code of Conduct:**

*I **KARNAM HARSHITH /192511384** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**

KARNAM HARSHITH

**RUBRICS FOR CALCULATION:**

| Criteria                  | Weightage | Level 4<br>(Exemplary)                                       | Level 3<br>(Proficient)                | Level 2<br>(Developing)                      | Level 1<br>(Beginning)          |
|---------------------------|-----------|--------------------------------------------------------------|----------------------------------------|----------------------------------------------|---------------------------------|
| Concept Identification    | 30%       | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps     | Basic understanding with some misconceptions | Poor or incorrect understanding |
| Linkages                  | 25%       | Completely correct answer with no errors                     | Mostly correct with minor errors       | Partially correct with noticeable errors     | Incorrect or irrelevant answer  |
| Organization              | 20%       | Correctly applies concepts to solve the question/problem     | Applies concepts with minor mistakes   | Limited or partially correct application     | No or incorrect application     |
| Integration of Literature | 15%       | Covers all required parts of the question                    | Covers most parts with minor omissions | Covers only some parts                       | Incomplete or missing response  |
| Clarity                   | 10%       | Clear, concise, and well-organized answer                    | Understandable with minor issues       | Somewhat unclear or poorly structured        | Difficult to understand         |

**Q1. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.**

**Schema**

A schema is the overall structure or blueprint of a database. It defines how data is organized into tables, attributes, relationships, and constraints. A schema changes very rarely.

**Example:**

A Student table with attributes:

**Student\_ID**

**Name**

**Department**

**Age**

This table design is called the database schema.

**Instance**

An instance is the actual data stored in the database at a particular moment. It changes whenever records are inserted, updated, or deleted.

**Example:**

| Student_ID | Name  | Department | Age |
|------------|-------|------------|-----|
| 101        | Rahul | CSE        | 19  |
| 102        | Priya | ECE        | 20  |

This data is an instance of the Student schema.

**Difference between Schema and Instance**

| Schema                | Instance           |
|-----------------------|--------------------|
| Blueprint of database | Actual data stored |
| Changes rarely        | Changes frequently |
| Defines structure     | Contains records   |

## **Types of Schema**

### **1. Logical Schema**

Describes the logical structure of the database.

Specifies tables, attributes, relationships, and constraints.

Independent of physical storage.

**Example:** Student(Student\_ID, Name, Department, Age)

### **2. Physical Schema**

Describes how data is physically stored.

Includes storage location, indexes, file organization, and access methods.

Example: Student records stored in B+ Tree index on SSD.

### **3. View Schema (External Schema)**

Shows only the required portion of the database to users.

Enhances security by hiding unnecessary data.

**Example:** A faculty member can view only Student\_ID, Name, and Department but not marks or fees.

**Q2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.**

### **Entities and Attributes**

#### **Customer**

- Customer\_ID (Primary Key)
- Name
- Address
- Phone
- Email

## **Account**

- Account\_No (Primary Key)
- Account\_Type
- Balance

## **Branch**

- Branch\_ID (Primary Key)
- Branch\_Name
- Location

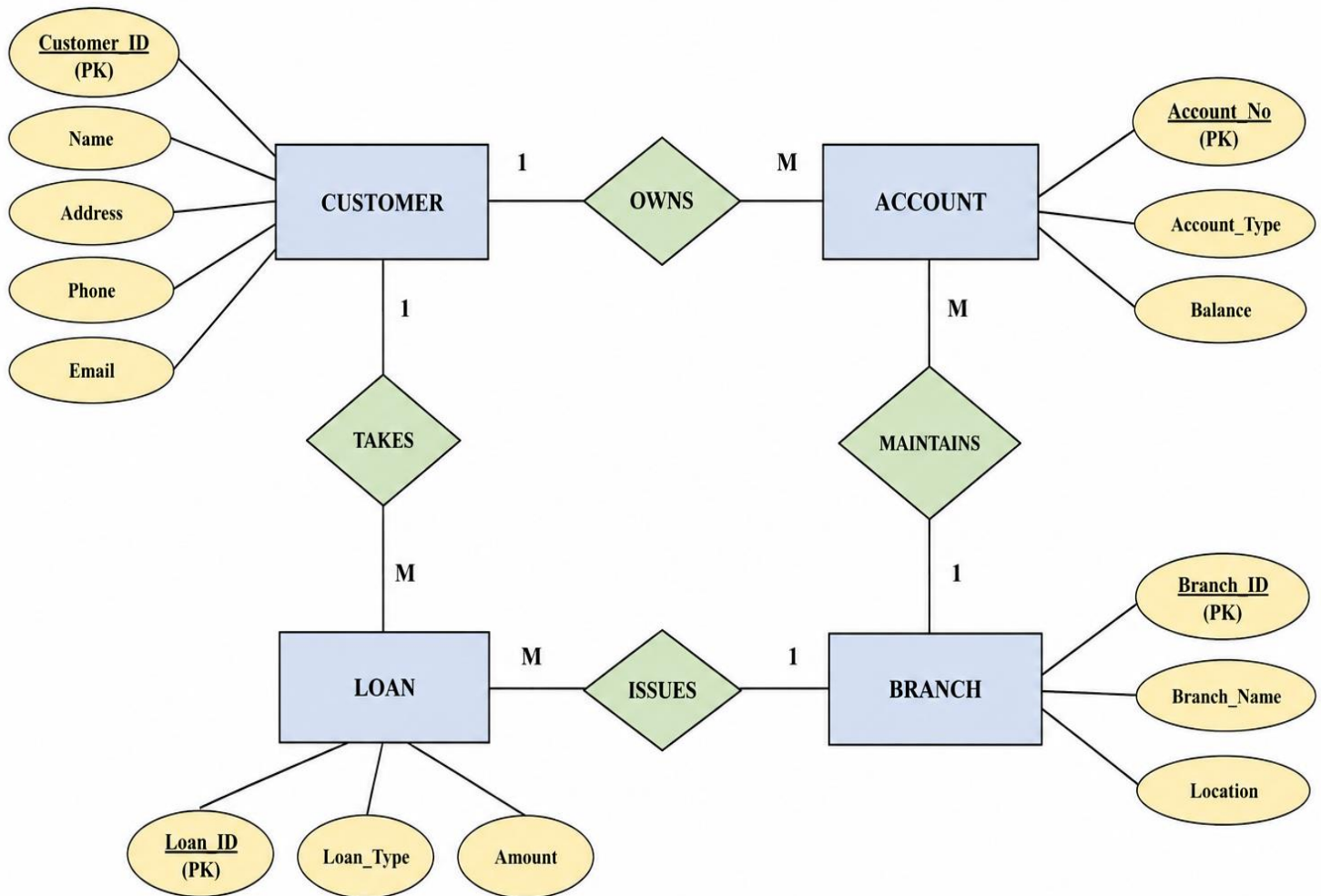
## **Loan**

- Loan\_ID (Primary Key)
- Loan\_Type
- Amount

## **Relationships**

- Customer owns Account (1:M)
- Branch maintains Account (1:M)
- Customer takes Loan (1:M)
- Branch issues Loan (1:M)

## BANK MANAGEMENT SYSTEM – ER DIAGRAM



**Q3. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.**

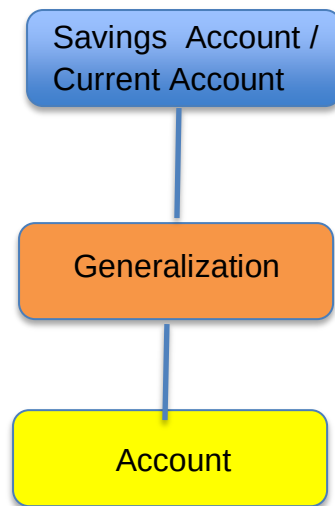
### **Extended ER (EER) Model**

The Extended Entity-Relationship (EER) Model is an enhancement of the ER model. It supports advanced concepts like inheritance, specialization, generalization, and aggregation to represent complex real-world applications.

#### **1. Generalization**

Generalization combines multiple lower-level entity types into a higher-level entity based on common characteristics.

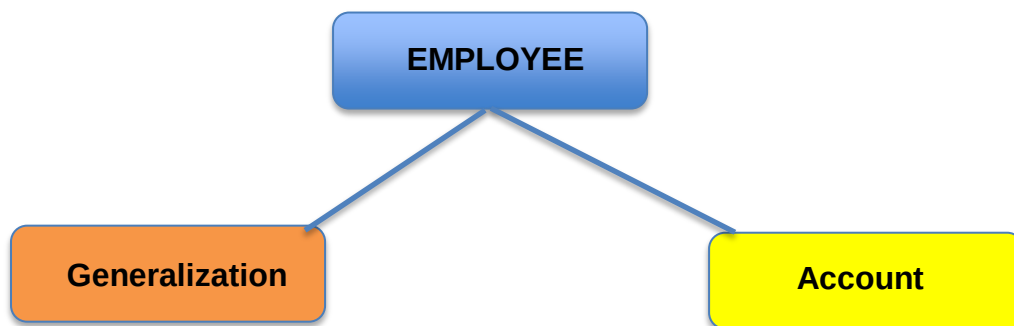
Example:



## 2. Specialization

Specialization divides a higher-level entity into lower-level entities based on distinguishing characteristics.

**Example:**



Both Doctor and Nurse inherit Employee attributes.

## 3. Aggregation

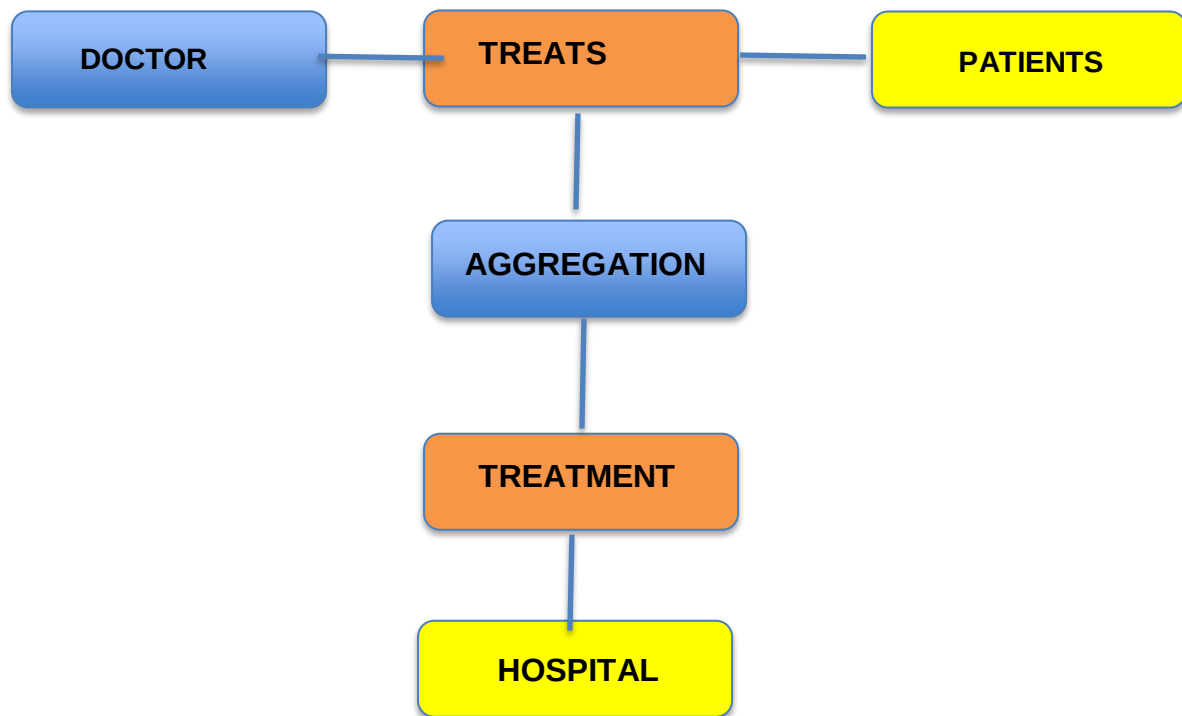
Aggregation treats a relationship as a higher-level entity so that it can participate in another relationship.

**Example:**

A Doctor treats a Patient.

This "Treats" relationship can be aggregated into a new entity called Treatment, which is then related to Hospital.

Doctor ---- Treats ----



**Q4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.**

### **Strong Entities**

#### **Patient**

- Patient\_ID (PK)
- Name
- Age
- Gender

#### **Doctor**

- Doctor\_ID (PK)
- Name



- Specialization

## **Hospital**

- Hospital\_ID (PK)
- Name
- Address.

## **Weak Entity**

### **Dependent**

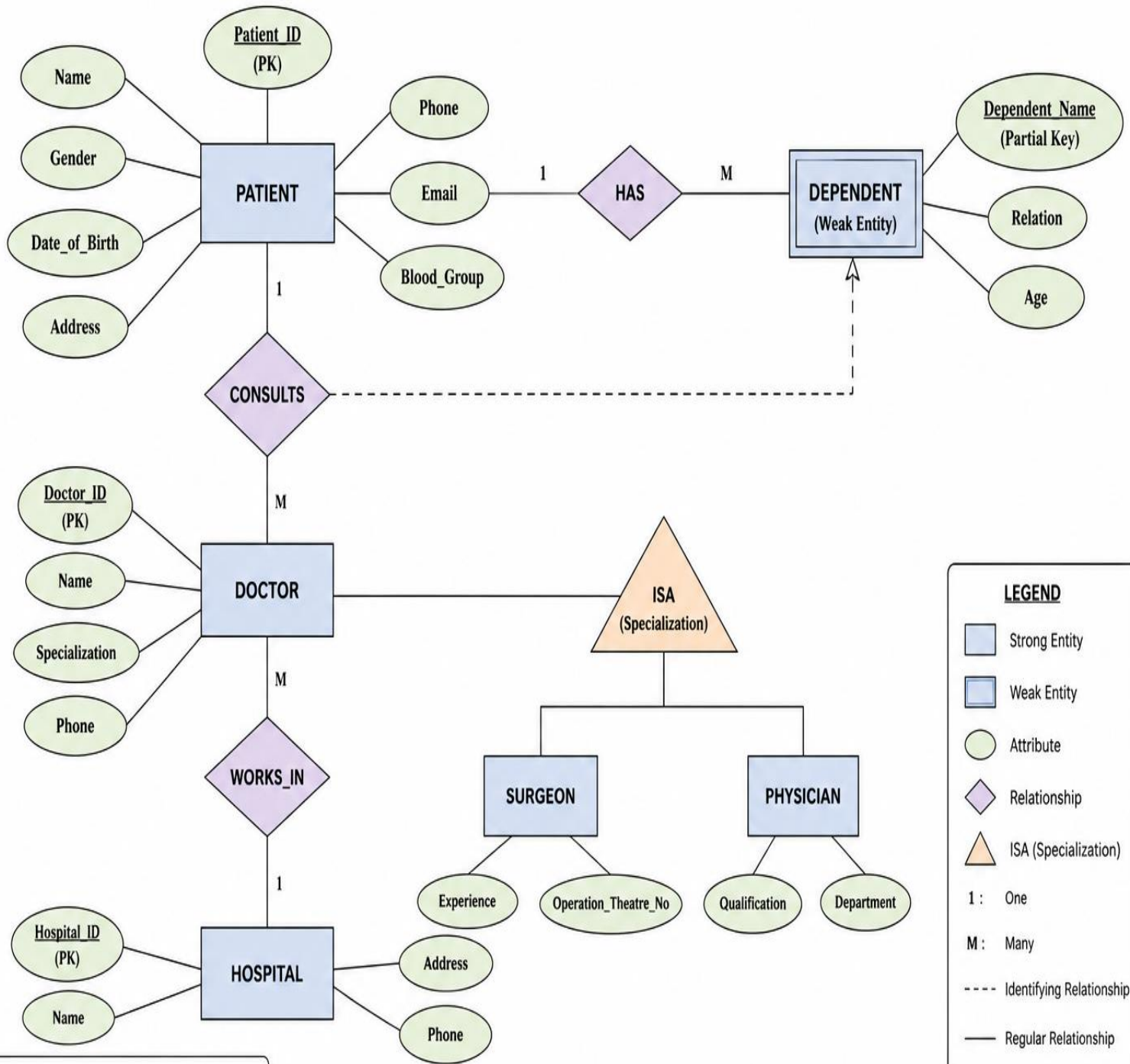
- Dependent\_Name (Partial Key)
- Relation
- Age

Dependent cannot exist without Patient.

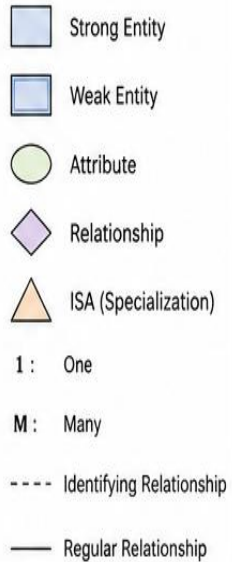
## **Relationships**

- Patient consults Doctor
- Doctor works in Hospital
- Patient has Dependent (Weak Entity)

# EER DIAGRAM FOR HOSPITAL MANAGEMENT SYSTEM



## LEGEND



## Notes:

- **DEPENDENT** is a weak entity and cannot exist without **PATIENT**.
- **DOCTOR** is specialized into **SURGEON** and **PHYSICIAN** (ISA hierarchy).

**Q5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.**

**Database Administrator (DBA)**

A Database Administrator (DBA) is responsible for managing, maintaining, securing, and ensuring the efficient operation of databases in an organization.

**Responsibilities**

1. Install and configure DBMS software.
2. Create and maintain databases.
3. Manage users and assign privileges.
4. Ensure data security and privacy.
5. Perform backup and recovery operations.
6. Monitor database performance and optimize queries.
7. Maintain data integrity and consistency.
8. Handle concurrency control and transaction management.
9. Troubleshoot database issues.
10. Upgrade and patch database systems.

**Conclusion**

A DBA ensures that databases are secure, reliable, efficient, available, and recoverable, enabling smooth operation of organizational applications.

**THANK YOU**